

BISHA, Saudi Arabia

WMO: 410840

Lat: 19.984N

Lon: 42.621E

Elev: 3887

StdP: 12.75

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	44.6	47.9	8.6	9.8	66.5	13.8	12.8	70.8	18.3	73.1	16.1	71.4	1.2	180	0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.4	104.4	64.4	103.9	64.1	102.5	63.6	70.7	88.2	69.4	88.1	68.2	88.7	8.2	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	111.3	76.3	64.3	104.3	76.3	62.6	98.1	75.8	37.2	88.0	35.9	87.7	34.9	88.4	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.0	15.5	13.5	DB	38.6	107.0	3.4	1.0	36.1	107.8	34.1	108.3	32.2	108.9	29.7	109.6	(4)
(5)			WB	29.2	73.0	3.6	1.8	26.6	74.3	24.5	75.3	22.4	76.3	19.8	77.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.5	64.1	68.8	74.5	79.5	85.2	89.2	91.1	90.7	85.6	77.0	70.9	64.7	(6)	
	DBStd	10.26	4.73	4.76	4.65	4.33	3.64	2.73	2.76	2.43	3.41	3.08	3.48	4.17	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	156	74	18	2	1	0	0	0	0	0	0	4	57	(9)	
	CDD50	10400	436	526	759	884	1092	1177	1275	1262	1069	837	628	455	(10)	
	CDD65	5082	44	125	297	435	627	727	810	797	619	372	182	47	(11)	
	CDH74	74434	783	1594	3376	5118	8889	11800	13159	12802	9275	4812	2059	767	(12)	
	CDH80	44549	175	555	1497	2689	5374	7823	8843	8543	5813	2432	677	128	(13)	
(14) Wind	WSAvg	5.0	4.1	4.8	5.5	5.5	5.1	5.0	5.6	5.3	5.2	5.9	4.4	3.8	(14)	
(15) Precipitation	PrecAvg	3.5	0.4	0.1	0.4	1.3	0.5	0.1	0.1	0.1	0.0	0.1	0.3	0.1	(15)	
	PrecMax	10.9	3.7	0.4	2.3	5.9	3.7	1.3	1.3	0.5	0.7	0.9	1.5	0.9	(16)	
	PrecMin	1.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	2.3	0.7	0.1	0.6	1.3	0.7	0.3	0.2	0.1	0.1	0.2	0.4	0.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.1	91.8	95.9	98.7	102.5	105.8	106.2	106.0	104.1	96.6	90.0	86.2	(19)	
		MCWB	59.3	60.9	63.1	63.0	63.7	64.5	65.6	65.7	64.7	61.4	61.1	59.8	(20)	
	2%	DB	84.3	89.4	93.1	96.8	100.8	104.2	104.4	104.3	102.1	94.5	88.1	83.3	(21)	
		MCWB	59.6	60.8	61.8	62.8	63.7	64.3	64.9	63.5	60.4	60.3	59.7		(22)	
	5%	DB	81.5	86.3	91.0	94.8	100.0	102.6	103.8	103.7	100.4	92.6	86.3	80.8	(23)	
		MCWB	59.2	60.4	62.4	62.8	63.6	63.1	63.9	64.5	62.6	59.5	60.1	58.9	(24)	
	10%	DB	78.7	83.8	88.0	92.8	98.1	101.9	102.3	102.2	98.7	91.0	84.3	78.7	(25)	
		MCWB	58.2	59.6	61.7	62.8	63.2	62.8	63.6	63.8	61.8	59.1	59.6	58.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.8	66.5	68.6	70.1	70.8	70.0	72.0	72.8	69.1	68.6	67.1	66.3	(27)	
		MCDB	80.2	81.5	87.0	85.0	86.6	90.1	91.3	88.9	91.3	78.8	79.8	80.2	(28)	
	2%	WB	63.1	64.6	67.1	68.6	69.2	67.8	69.9	71.1	66.7	65.9	65.2	64.2	(29)	
		MCDB	78.1	82.0	84.7	84.2	85.6	93.3	90.0	88.3	93.4	80.0	79.2	77.3	(30)	
	5%	WB	61.6	63.3	65.8	67.4	67.9	66.5	68.0	69.5	65.2	64.0	63.9	62.4	(31)	
		MCDB	76.5	80.8	82.7	83.2	85.6	94.2	93.0	89.7	93.8	81.4	78.7	75.8	(32)	
	10%	WB	60.0	61.8	64.4	66.3	66.7	65.3	66.5	67.7	63.9	61.8	62.6	60.3	(33)	
		MCDB	75.1	79.0	81.6	82.5	86.1	94.5	94.4	92.7	94.0	84.6	77.6	74.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	28.6	28.8	27.2	25.2	25.2	26.0	23.4	24.4	27.3	27.9	27.0	28.6	(35)	
		MCDBR	32.4	31.9	29.5	27.3	25.8	26.3	23.9	24.9	27.3	28.6	29.2	30.7	(36)	
	5% WB	MCWBR	15.9	14.7	12.5	9.9	8.8	9.5	9.3	9.7	11.0	12.4	13.6	15.4	(37)	
		MCWBR	28.1	28.2	25.9	24.5	24.6	24.9	22.7	23.3	25.1	24.4	24.6	26.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.379	0.414	0.464	0.512	0.557	0.585	0.623	0.629	0.531	0.421	0.405	0.376	(40)	
		taud	2.221	2.084	1.949	1.856	1.732	1.652	1.617	1.602	1.766	2.045	2.118	2.212	(41)	
	Ebn at Noon		283	278	269	256	243	234	225	225	246	272	271	279	(42)	
		Edh at Noon	42	50	60	67	75	80	83	85	71	52	46	41	(43)	
(44) All-Sky Solar Radiation	RadAvg		1590	1835	2065	2160	2313	2399	2250	2169	2139	2006	1679	1537	(44)	
	RadStd		73	74	85	110	112	65	80	55	55	56	64	72	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	+1.82	+2.11	N/A	N/A	N/A	N/A	(46)
(47)	Regional (3 neighbors)	+0.86	N/A	N/A	+1.03	+0.58	+0.38	N/A	-131	+360	+227	(47)

Nomenclature: See separate page